MIL-F-36380A
27 April 1972
SUPERSEDING
MIL-F-36380(DM)
11 February 1965

MILITARY SPECIFICATION

FITTING SET, CHEMICAL-BIOLOGICAL MASK OPTICAL INSERT

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a protective mask optical insert fitting set.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-C-586	Copper-Nickel-Zinc Alloy; Rod, Shapes, and Flat Products with Finished Edges (Flat Wire, Strip, and Bar).
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

FSC 6540

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STANDARDS

Federal

Fed. Test Method Std. No. 151	Metals; Test Methods.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center (DPSC)

20325	Insert, Optical, Field Protective Mask.

(Copies of specifications, standards, and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Components. The fitting set shall consist of the following components:

(a) Chemical-Biological Mask optical insert. One of each of following CB Mask optical inserts (see stock number listed below), fitted with plastic lenses which shall be marked on the surface with a vertical and a horizontal line through the geometric center, extending the full distance on the lens, with five (5) 1-millimeter calibrations along the vertical and horizontal lines on each side of the geometric center to permit calculation of decentration required in prescription lenses. The length of the calibration lines shall be 2-millimeters:

Stock No.	Item Identification	Position	Bridge
6540-656-1240	Insert, Optical, Chemical-Biological Mask	Lower	Wide
6540-656-1243	Insert, Optical, Chemical-Biological Mask	Central	Narrow
6540-656-1244	Insert, Optical, Chemical-Biological Mask	Lower	Narrow
6540-656-1247	Insert, Optical, Chemical-Biological Mask	Central	Wide
6540-656-1248	Insert, Optical, Chemical-Biological Mask	Upper	Narrow
6540-656-1251	Insert, Optical, Chemical-Biological Mask	Upper	Wide

The above protective mask optical inserts shall conform to the Defense Medical Purchase Description of the issue in effect on date of invitation for bids, except for packaging and packing, for each of the items listed above.

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) Accessories.

(1) One bottle of 100 temporal strut screws, No. 1-72 by 1/16 inch long, slotted setscrew, corrosion-resisting steel, type 302 or 303, of a temper suitable for the purpose intended. Screws shall be the same as temporal strut screws used in the above inserts.

(2) One bottle of 100 eyewire screws, 0.050-90 by 3/16 inch long, optical screw, nickel silver, conforming to alloy No. 752, as specified in table I of QQ-C-586, of a temper suitable for the purpose intended. Screws shall be the same as the eye wire screws used in the above inserts.

(3) Two screw taps for the temporal strut screws.

(4) Two screw taps for the eyewire screws.

(5) One screwdriver, optician.

:) Case. The case shall be of the minimum size required to accommodate protective mask optical inserts and accessories listed above. The frame . be made of white pine or other suitable wood, with laminated fibre top ottom, and flannel or velvet lined inside lid and sides. The case shall six insert compartments, inclined for maximum visibility, with fibre le set in slotted wooden endpieces. The case shall be covered with it leather, with a 1-1/2 inch flap extending the entire length of the and fastened with a one-button snap fastener. The color of the imitation ier and lining shall be black. As an alternate, the case shall be fabricated suitably durable plastic such as polyvinyl chloride or acrylonitrile buta- e styrene one-piece, or with molded insert containing required compartments. shall be black.

.2 Materials. Except as specified herein, materials shall conform to the icable portions of the specifications, standards, and drawing referenced herein, shall be capable of meeting the requirements specified therein. In addition, metals shall conform to the chemical requirements of their applicable speci- ions when tested as specified in 4.4.1. When materials to be used are not ific ally designated, those used shall be of the highest commercial quality entirely suitable for the purpose intended.

.2.1 Plastic for lenses. Lenses shall be made of slow burning, clear, less, transparent cellulose acetate, suitable for etching. As an alternate, es may be made of a thermosetting optical polymer produced from the poly- zation of alkyl diglycol carbonate, known to the trade as CR-39. Lenses l be clean and substantially free from superficial undulations, striae, bubbles, s, scratches, opaque particles, or any other imperfections which detract from r appearance or may impair their serviceability.

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3.3 Construction of plastic lenses. The plastic lens shall be a meniscus lens having a 6 diopter base curve, shall conform to the requirements of the drawing, and shall fit properly in the binocular lens subassembly of the same drawing. Lenses shall be marked as specified in 3.1.

3.3.1 Thickness of plastic lens. The thickness of the plastic lens shall be 0.050 $\pm$ 0.10 if constructed from clear cellulose acetate or 0.085 $\pm$ 0.10 if constructed of CR-39.

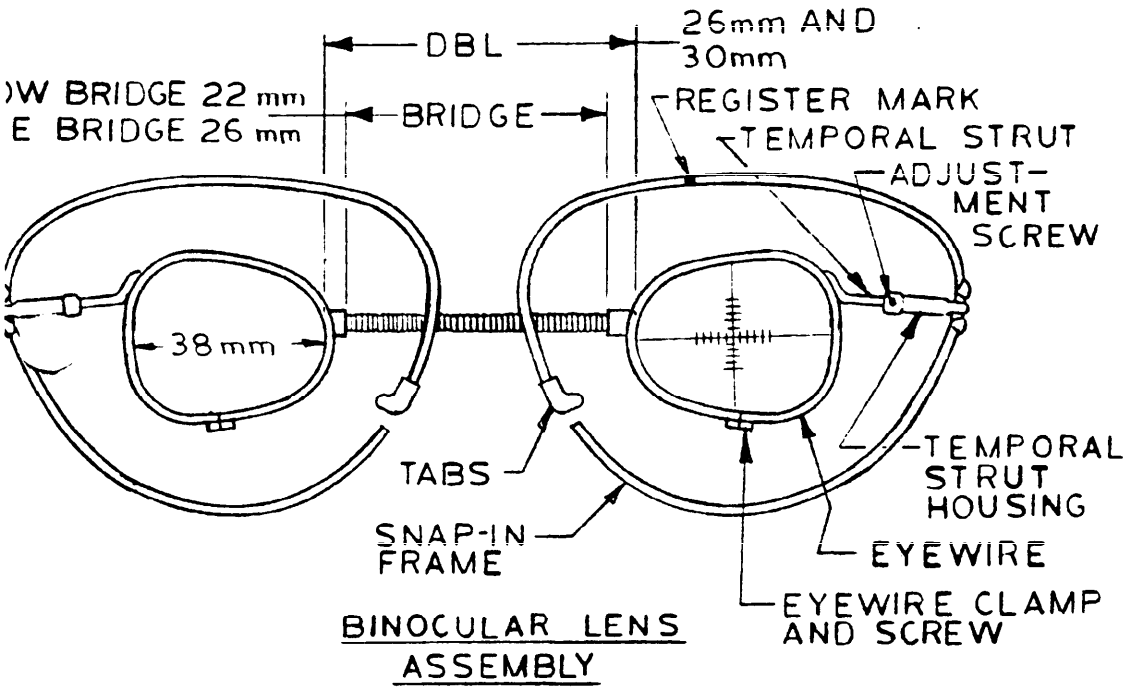
3.4 Instructions. One set of instructions shall be inserted with each fitting set. The instructions shall contain the information specified as follows:

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INSTRUCTIONS
FOR
FITTING OPTICAL INSERTS IN M-17 MILITARY PROTECTIVE MASK

The M-17 Optical Insert Fitting Kit contains six (6) optical inserts
lows: 3-wide bridge inserts with lenses in upper, lower, and central
ons, 3-narrow bridge inserts with lenses in upper, lower, and central
ons. When inserts are ordered, the bridge width and lens position will
ered under the "Detail" section of DD Form 771 (Spectacle Order Form).

Diagram and Nomenclature of the M-17 Insert



Method for Attaching Insert.

a. Snap the headstrap of the mask over the front of the facepiece and
ack the edges of the mask so that you can get to the inside of the mask
ces easily.

b. Grasp the plastic covered tabs on the frames and insert the lower
n of the frames into the channel in the lower section of the mask eyepiece,
the upper section down and into the upper channel of the mask eyepiece.
gister mark (small dot) on the insert frame should coincide with the mark
mask eyepiece that indicates its vertical center line.

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c. Repeat the same procedure for the other snap-in frame.

d. To remove the insert, grasp the plastic covered tabs and pull the upper section of the frame down and out of the mask eyepiece channel. Remove each frame assembly individually.

e. Caution should be taken in handling the insert to keep fingerprint smudges from the insert lenses and inner surface of the eyepiece.

4. Method for Selecting Proper Insert.

a. Select the narrow or wide bridge insert with the central position lenses and attach as set forth in paragraph 3 above. Have the wearer put on the mask and adjust it properly, which means primarily seeing to it that the eyepieces are well centered, both horizontally and vertically. This is particularly important because many men wear their masks off-center and, frequently, with one eyepiece higher than the other.

b. Have the wearer direct his gaze straight ahead and check for correctness of bridge width and lens position. If center of pupil appears lower than the center of the insert, the lower position insert would be the correct one.

c. After carrying out the procedure in paragraph b, above select the proper insert and attach it to the wearers mask. Have the wearer adjust the mask and make final check before ordering.

5. Decentration of lenses using the M-17 Optical Insert is the same as when utilizing regular spectacles. The geometric center of the wide bridge insert is 68 mm and the narrow bridge insert 64 mm. As an example, if an individual was fitted with the narrow bridge insert and the individual had a distant pupillary measurement (Pd) of 66 mm, then the decentration would be out 1 mm for each lens.

6. The desired pantoscopic tilt and vertex distance adjustment may be achieved by first loosening the set screws in the temporal strut housings and removing the Binocular Lens Assembly. To obtain the correct pantoscopic tilt, grasp the temporal struts close to the eyewires with opticians adjustment pliers, then bend the struts to the required angle. The vertex distance may be controlled by inserting the temporal struts into the temporal strut housings the required depth and then firmly tightening the set screws.

7. It is recommended that the measurement, fitting, and demonstrations of insertion and removal of the M-17 Optical Insert be accomplished by an Optometry Officer or enlisted Optical Laboratory Specialist (MOS 453).

8. Optical inserts for military protective masks are considered personal items and accompany the individual when transferred.

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3.5 Workmanship. Workmanship shall be first class throughout. Fitting ts shall be free from defects which detract from their appearance or may pair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract purchase order, the supplier is responsible for the performance of all in- ection requirements as specified herein. Except as otherwise specified in e contract or order, the supplier may use his own or any other facilities itable for the performance of the inspection requirements specified herein, less disapproved by the Government. The Government reserves the right to rform any of the inspections set forth in the specification where such in- ections are deemed necessary to assure supplies and services conform to pre- ribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the ntractor shall be maintained by the contractor and made available to the Gov- nment, upon the Government's request, at any time, or from time to time, during e performance of the contract and for a period of 3 years after delivery of the pplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as th examination (such as visual and auditory investigation without the use of ecial laboratory appliances or procedures) and testing (determination by chr·al means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the nufacturer of the inserts, screws, and lenses, may be furnished in lieu of tual performance of such testing by the contractor, provided lot identity s been maintained and can be demonstrated to the Government. The certifi- te shall include the name of the contractor, the contract number, the name the manufacturer or supplier, the name of the component/material, the lot mber, and test results.

4.2 Sampling.

4.2.1 Sampling for examination. Sampling for examination shall be con- cted in accordance with MIL-STD-105. The inspection levels and acceptable ality levels shall be as indicated in table I. Unit of product for exam- ation purposes shall be one fitting set.

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Table I. Sampling Plan

For visual examination	Inspection level	AQL (percent defective)
Major A	II	1.0
Major B	S-3	1.0
Minor A	II	4.0
Minor B	S-3	4.0

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 as follows:

	Inspection level	AQL (percent defective)	Minimum sample size
Component material testing	S-1	*	3

*Acceptance number shall be zero (0) for all sample sizes.

4.2.3 For metals. Four ounces of each lot of metal used in the construction of the fitting set shall be selected for chemical analysis.

4.2.4 Component materials. Sufficient samples of the component materials shall be selected from each lot of material used in the manufacture of the fitting sets to perform tests as required by the applicable subsidiary specifications or by tests specified herein. When sample size for component materials covered by the subsidiary specifications is not specified, the sample size for each test shall be as indicated in 4.2.2. The unit of product for testing shall be the quantity necessary to perform all required tests one time each taking into consideration the fact that where possible, the same material may be used to perform two or more tests.

4.3 Examination. The fitting sets shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with, but not restricted to, the following classification of defects. Additional defects shall be classified in accordance with MIL-STD-105.

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Categories and defects

Major A		
	101	Set is not proper design.
	102	Component missing.
	103	Accessory missing (total amount required).
Major B		
	150	Dimension not in tolerance.
	151	Case or components damaged.
	152	Case unlined.
	153	Lenses not calibrated or improperly calibrated.
	154	Instruction sheet missing or incomplete.
Minor A		
	201	Metal parts not free of burrs and sharp edges.
	202	Case not uncovered or improperly covered.
	203	Snap fastener missing.
Minor B		
	250	Wrong color case or lining.
	251	Improper fastener used.
	252	Case is substantially above minimum required size.

4.4 Tests. Necessary tests shall be conducted to determine compliance with specification requirements.

4.4.1 Chemical analysis of metals. The samples selected in accordance with 4.2.3 shall be anlayzed in conformance with method 111 or 112 of Federal Test Method Standard 151. In case of dispute, method 111 shall be used for referee analysis.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One fitting set, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table II shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

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Table II. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	Not Required	6 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Each unit shall be packaged in a box of appropriate size constructed in accordance with PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent spilling of contents under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Special requirements as specified in PPP-B-636 shall apply for fiberboard boxes.

5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed in accordance with 30.4 of PPP-B-636.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4, of PPP-B-636.

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4 Strapping. Strapping, when required, shall be in accordance with of applicable box specification.

Strapping shall not be required for shipments forwarded to a activity within the continental limits of the United States for nd redistribution.

Level B.

1 Exterior container. Exterior container shall be designed for a ad and constructed in accordance with PPP-B-585, class 1, style 3; ., domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. hall be as specified in the appendix of the applicable box specifi- Fiberboard box shall be closed by method II of PPP-B-636. In addition, d boxes shall conform to the special requirements specified in .

Level C. The subject commodity shall be packed in substantial com- containers of the type, size, and kind commonly used for the purpose, ucted as to insure acceptance and safe delivery by common or other , at the lowest rate, to point of delivery called for in the contract se order.

ng.

Unit packages. Each unit package shall be marked in accordance with 129. When labels are utilized, waterproofing shall be required only licable carton is fabricated of water-resistant material. Contract se order number and name of contractor shall be shown.

Exterior container. Exterior container shall be marked in accordance -STD-129.

OTES

Intended use. The fitting set covered herein is intended to provide determining the appropriate mask insert for personnel requiring visual on.

Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Federal Stock Number (FSN).
(c) Selection of applicable levels of packing (see 5.2).

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6.3 The fitting set covered by this specification is listed in the Federal Supply Catalog under the following stock number:

Federal Stock Number	Item Identification
6540-656-1252	FITTING SET, CHEMICAL-BIOLOGICAL MASK OPTICAL INSERT

6.4 This specification does not include all types, classes, grades sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Military requirements.

6.5 Supersession Data. This specification includes the requirements of MIL-F-36380(DM), dated 11 February 1965.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6540-1026

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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NDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval No. 22-R255

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NT IDENTIFIER AND TITLE

FSN: 6540-656-1252

F ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT

IVE PARAGRAPH NUMBER AND WORDING.

ECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

MENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

HE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

ARKS

TTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

FORM 1426 72

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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